

A CNN-Based Bird Species Identification System for Biodiversity Monitoring Using Transfer Learning

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Abstract—Accurate identification of bird species from photographs supports biodiversity monitoring, ecological research and citizen-science programmes, but it is a fine-grained recognition problem: many species differ only in subtle plumage details, while pose, lighting and background vary widely within a species. This paper presents a bird species identification system based on convolutional neural networks (CNNs) and transfer learning. Three ImageNet-pretrained backbones, VGG16, ResNet50 and EfficientNet-B0, are trained under an identical protocol: a regularised classification head (global average pooling, a fully connected layer with batch normalisation and dropout, and a 200-way softmax), data augmentation, and two-stage training in which the backbone is first frozen and its top layers are then fine-tuned. EfficientNet-B0 achieves the highest validation accuracy while being the smallest and fastest model, and is selected for the system. On the official test split of CUB-200-2011 (5,794 images, 200 species), evaluated without bounding-box or part annotations, the system reaches a top-1 accuracy of 73.1%, a top-5 accuracy of 92.8% and a macro F1-score of 73.2%, compared with 70.4% and 68.8% top-1 for ResNet50 and VGG16. Ablations show that fine-tuning improves every backbone, whereas the data augmentation used does not improve accuracy. We further analyse per-species errors, report model size and inference time, and use Grad-CAM to show which image regions drive each prediction.

Index Terms—Bird species classification, fine-grained image recognition, convolutional neural networks, transfer learning, EfficientNet, Grad-CAM, biodiversity monitoring

I. INTRODUCTION

Identifying bird species is a fundamental task in biodiversity research, ecological monitoring and conservation. Reliable species records support the estimation of population trends, the study of migration routes and the evaluation of conservation interventions. Traditionally this work depends on expert ornithologists and field observation, which is labour-intensive, prone to observer bias and difficult to scale to the volume of images now produced by camera traps, field surveys and citizen-science platforms.

Deep learning, and convolutional neural networks (CNNs) in particular, has transformed image recognition by learning visual features directly from data instead of relying on hand-designed descriptors [1], [2]. Bird species identification is nevertheless harder than generic object recognition. It is a

fine-grained problem: inter-class differences are small (closely related warblers or sparrows may differ only in a wing bar or eye ring), while intra-class variation is large because of pose, viewpoint, illumination, occlusion, background clutter and differences between male, female and juvenile plumage [3]. In addition, labelled data are limited: the widely used CUB-200-2011 benchmark provides only about 30 training images per species [3].

Transfer learning addresses the data limitation by reusing features learned on ImageNet [4] and adapting them to the target domain [5]. Many ImageNet backbones are available, however, and they differ widely in accuracy, size and computational cost, which matters when a system is to be deployed on phones or field devices. This paper presents a complete bird species identification system built on transfer learning and evaluates it carefully. Rather than proposing a new architecture, we compare three widely used backbones under identical conditions, build the system on the one that performs best on held-out validation data, and report every result on the official test split, together with the ablations and error analysis needed to interpret it.

The main contributions of this paper are as follows:

- We design a bird species identification system that couples an ImageNet-pretrained EfficientNet-B0 backbone with a regularised classification head and a two-stage transfer-learning strategy (frozen backbone, then partial fine-tuning), using only image-level labels.
- We compare VGG16, ResNet50 and EfficientNet-B0 under an identical training protocol on the official CUB-200-2011 split across all 200 species, and quantify the contribution of fine-tuning and data augmentation through ablations.
- We provide a detailed analysis of the system's behaviour: top-1 and top-5 accuracy, macro-averaged precision, recall and F1-score, ROC-AUC, per-species performance, the most frequently confused species pairs, model size and inference time, and Grad-CAM visual explanations of the predictions.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the dataset, model and training procedure. Section IV presents and discusses the results, and Section V concludes the paper.

II. RELATED WORK

A. Hand-crafted features

Early automated bird classification relied on hand-designed descriptors such as colour histograms, histograms of oriented gradients [6] and local binary patterns [7], combined with classical classifiers such as support vector machines [8], k -nearest neighbours and random forests. These methods laid the groundwork for automated species recognition, but their accuracy depended heavily on feature engineering and on consistent imaging conditions.

B. CNNs and transfer learning

CNNs learn hierarchical features directly from pixels [1]. VGG16 [2] stacks small 3×3 convolutions to a depth of 16 weight layers; ResNet [9] adds residual connections that make much deeper networks trainable; and EfficientNet [10] scales depth, width and input resolution jointly from a compact baseline built from mobile inverted bottleneck blocks [11], reaching similar accuracy with far fewer parameters and operations. Pretrained on ImageNet [4], these networks are routinely fine-tuned for domain-specific tasks. Yosinski *et al.* [5] showed that early layers learn general features while later layers become increasingly task-specific, which motivates freezing the early layers and fine-tuning the later ones when training data are scarce. Data augmentation is a standard complement that increases the effective size and diversity of the training set [12]. CNN feature extractors pretrained on ImageNet have likewise been adapted to other domain-specific image tasks, such as plant disease detection [13] and the classification of partially obscured food items with a modified VGG19 [14].

C. Fine-grained bird recognition

Methods designed specifically for fine-grained recognition localise and exploit discriminative parts. Part-based R-CNN [15] detects object parts and pools features from them; bilinear CNNs [16] model pairwise feature interactions; the recurrent attention CNN [17] progressively zooms into discriminative regions; and TransFG [18] uses a vision transformer that selects informative image patches. These methods achieve high accuracy on CUB-200-2011 but typically require higher input resolutions, more complex architectures, or more computation than a standard transfer-learning pipeline.

D. Datasets and ecological monitoring

Public benchmarks have been essential for progress: CUB-200-2011 [3], NABirds [19] and iNaturalist [20] provide labelled bird images at increasing scale. Beyond images, acoustic monitoring is widely used; BirdNET [21] identifies birds from their vocalisations, and Gómez-Gómez *et al.* [22] compared large and small-footprint CNNs for audio-based classification with deployment on low-cost devices in mind.

BirdSAT [23] combines ground-level bird images with satellite imagery and metadata for both classification and species distribution mapping, and image-based deep learning is also used for wider environmental monitoring, for example to detect marine plastic waste in aerial and satellite images [24].

E. Interpretability

Ecologists need to understand why a model makes a prediction before trusting it. Gradient-weighted class activation mapping (Grad-CAM) [25], LIME [26] and SHAP [27] highlight the input regions or features responsible for a decision. Kumar and Kondaveeti [28] applied transfer learning with LIME to CUB-200-2011 and measured how well the explanations overlap the bird. Following this line, we use Grad-CAM to check whether our system attends to the bird rather than to the background.

III. METHODOLOGY

A. System Overview

Fig. 1 summarises the system. A user supplies a photograph of a bird. The image is resized, and during training it is also randomly augmented. The EfficientNet-B0 backbone extracts convolutional feature maps, which the classification head converts into a probability distribution over the 200 species. The system returns the most likely species with their confidence scores, and a Grad-CAM heat map that shows which regions of the image supported the prediction.

B. Dataset

We use the Caltech-UCSD Birds-200-2011 (CUB-200-2011) dataset [3], which contains 11,788 images of 200 North American bird species. We use the official split of 5,994 training and 5,794 test images, which makes our results directly comparable with published work. From the official training images we hold out a stratified 10% (the same fraction of every species) as a validation set, which is used for early stopping, learning-rate scheduling and model selection. This gives 5,394 training, 600 validation and 5,794 test images. The test set is used only for the final evaluation. Only image-level species labels are used; the bounding boxes and part annotations supplied with the dataset are not used for training or testing.

C. Preprocessing and Data Augmentation

All images are resized to 224×224 pixels, the input resolution of all three backbones, and normalised exactly as during each backbone's ImageNet pretraining (for EfficientNet-B0 this normalisation is built into the network). To improve generalisation and reduce the effect of variations in viewpoint, scale and lighting, the following random transformations are applied to every training image (and never to validation or test images) [12]:

- horizontal flipping, since birds face left and right equally often;
- rotation within $\pm 15^\circ$, to simulate natural variation in orientation;

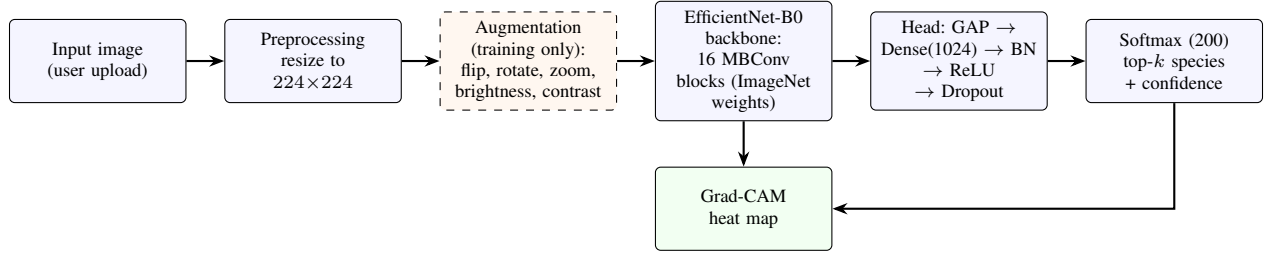


Fig. 1. Overview of the proposed bird species identification system. The dashed stage is applied only during training. Grad-CAM combines the gradients of the predicted class with the last convolutional feature maps to explain each prediction.

- zooming by up to $\pm 20\%$, to provide robustness to distance and cropping;
- brightness and contrast changes of up to $\pm 20\%$, to cope with varying illumination.

D. Model Architecture

The backbone of the proposed system is EfficientNet-B0 [10] pretrained on ImageNet, with its original classification layer removed. It begins with a 3×3 convolution followed by 16 mobile inverted bottleneck (MBConv) blocks [11] arranged in seven stages, and ends with a 1×1 convolution to 1280 channels. Each MBConv block expands its input with a 1×1 convolution, applies a depthwise 3×3 or 5×5 convolution, recalibrates the channels with a squeeze-and-excitation unit [29], and projects back with a 1×1 convolution, with a residual connection when the input and output shapes match. All convolutions are followed by batch normalisation and, except for the projections, by the Swish activation [30]. For a 224×224 input the backbone produces a $7 \times 7 \times 1280$ feature map.

The classification head consists of:

- global average pooling (GAP), which reduces the feature map to a 1280-dimensional vector. Flattening would instead produce a 62,720-dimensional vector and over 64 million weights in the next layer; GAP keeps the head small and less prone to overfitting;
- a fully connected layer with 1024 units, followed by batch normalisation [31] and ReLU activation;
- dropout [32] with rate 0.5;
- a fully connected output layer with 200 units and softmax activation.

Table I lists the parameters of each part; the complete model has 5.6M parameters. The VGG16 [2] and ResNet50 [9] models used for comparison have exactly the same head on top of their own backbones, whose final feature maps have 512 and 2048 channels.

E. Two-Stage Training

Training proceeds in two stages, following the observation that pretrained low-level features transfer well while high-level features benefit from adaptation [5]:

- 1) *Stage 1 (frozen backbone)*. All backbone weights are frozen and only the head (1.52M trainable parameters for

TABLE I
ARCHITECTURE OF THE PROPOSED MODEL

Layer(s)	Output shape	Parameters
EfficientNet-B0 backbone (stem conv, 16 MBConv blocks, 1×1 conv)	$7 \times 7 \times 1280$	4,049,571
Global average pooling	1280	0
Dense + batch norm. + ReLU	1024	1,315,840
Dropout (0.5)	1024	0
Dense + softmax	200	205,000
Total		5,570,411

EfficientNet-B0) is trained with the Adam optimiser [33] at a learning rate of 10^{-3} .

- 2) *Stage 2 (fine-tuning)*. The top of the backbone is unfrozen and the whole network is trained further at a learning rate of 10^{-4} . For EfficientNet-B0 this comprises stages 5–7 and the final 1×1 convolution; for VGG16, convolutional blocks 4 and 5; and for ResNet50, stages conv4 and conv5. Batch-normalisation layers inside the backbone remain frozen, so that their ImageNet statistics are not disturbed by the small mini-batches.

In both stages the training is stopped when the validation loss has not improved for 5 epochs, and the weights from the epoch with the lowest validation loss are restored. The learning rate is halved when the validation loss stops improving for 2 epochs. Table II summarises these settings. For EfficientNet-B0, stage 1 lasted 17 epochs and stage 2 lasted 12 epochs. The stage-2 setting was chosen on the validation set: in a preliminary experiment with VGG16, fine-tuning only the last block at 10^{-5} gave a validation accuracy of 62.5%, compared with 67.8% for the setting above, which we then used for all models.

F. Backbone Selection

VGG16, ResNet50 and EfficientNet-B0 are trained with identical data, preprocessing, augmentation, head and two-stage schedule. The backbone used in the system is the one with the highest top-1 accuracy on the validation set, so that the test set plays no part in the choice. The validation accuracies are 67.8% for VGG16, 72.8% for ResNet50 and 75.2% for EfficientNet-B0, which is therefore selected.

TABLE II
TRAINING HYPERPARAMETERS (IDENTICAL FOR ALL BACKBONES)

Hyperparameter	Stage 1	Stage 2
Trainable layers	head only	head + top of backbone
Optimiser	Adam	Adam
Initial learning rate	10^{-3}	10^{-4}
Batch size	32	32
Maximum epochs	30	30
Early stopping	patience 5	patience 5
LR reduction	$\times 0.5$, patience 2	$\times 0.5$, patience 2
Loss	categorical cross-entropy	

G. Mathematical Formulation

A standard convolutional layer computes feature maps

$$Z = (X * W) + b, \quad (1)$$

where X is the input, W a set of learnable kernels and b the bias. The depthwise convolution inside an MBConv block applies one kernel K_c to each input channel separately, $Z_c = X_c * K_c$, and is followed by a pointwise 1×1 convolution that mixes the channels; this factorisation needs far fewer parameters and operations than a full convolution [11]. The squeeze-and-excitation unit [29] rescales each channel c by a learned gate,

$$s = \sigma(W_2 \delta(W_1 \text{GAP}(X))), \quad \tilde{X}_c = s_c X_c, \quad (2)$$

where σ is the sigmoid and δ the activation function. The backbone uses the Swish activation $f(x) = x \sigma(x)$ [30], and the head uses ReLU, $f(x) = \max(0, x)$. Batch normalisation standardises each activation using the mini-batch mean μ_B and variance σ_B^2 and then rescales it,

$$\hat{x} = \gamma \frac{x - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} + \beta, \quad (3)$$

where γ and β are learned. The softmax layer converts the output logits z_i into class probabilities over the $K = 200$ species,

$$\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}, \quad (4)$$

and the model is trained by minimising the categorical cross-entropy between the one-hot label y and the prediction $\hat{y}$,

$$L = - \sum_{i=1}^K y_i \log \hat{y}_i. \quad (5)$$

The parameters are updated with Adam [33], which combines momentum with per-parameter adaptive learning rates.

H. Evaluation Metrics

For a multi-class problem, the confusion-matrix counts are defined per species. For species k : a *true positive* (TP_k) is a test image of species k predicted as k ; a *false positive* (FP_k) is an image of another species predicted as k ; a *false negative* (FN_k) is an image of species k predicted as another species;

and a *true negative* (TN_k) is an image of another species not predicted as k . From these,

$$P_k = \frac{TP_k}{TP_k + FP_k}, \quad R_k = \frac{TP_k}{TP_k + FN_k}, \quad (6)$$

$$F1_k = 2 \frac{P_k R_k}{P_k + R_k}. \quad (7)$$

Macro-averaged precision, recall and F1-score are the means of P_k , R_k and $F1_k$ over all 200 species, so every species counts equally. Top-1 accuracy is the fraction of test images whose most probable predicted species is correct,

$$\text{Acc} = \frac{1}{N} \sum_{n=1}^N \mathbb{1} \left[\arg \max_k \hat{y}_{n,k} = c_n \right], \quad (8)$$

and top-5 accuracy counts a prediction as correct if the true species c_n is among the five most probable. We also report the macro-averaged one-vs-rest area under the ROC curve (AUC) and the mean average precision (mAP), which evaluate the ranking of the predicted probabilities rather than only the top prediction.

I. Grad-CAM Visual Explanations

To explain a prediction for class c , Grad-CAM [25] weights each feature map A^m of the last convolutional layer of the backbone (the $7 \times 7 \times 1280$ output of the final 1×1 convolution of EfficientNet-B0) by the spatially averaged gradient of the class score y^c ,

$$\alpha_m^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^m}, \quad L^c = \text{ReLU} \left(\sum_m \alpha_m^c A^m \right), \quad (9)$$

and the map L^c is upsampled to the image size and overlaid on the input.

IV. RESULTS AND DISCUSSION

A. Experimental Setup

All models were implemented in TensorFlow/Keras [34] and trained on a single Tesla T4 GPU with mixed-precision arithmetic. All reported numbers are computed on the 5,794 official test images and all 200 species. Inference time is the median time to classify a single image (batch size 1) on a 2-core Intel Xeon at 2.20GHz CPU, with all models timed in the same session, and model size is the storage needed for the weights in 32-bit floating point.

B. Backbone Comparison

Table III reports the validation and test performance of the three backbones and of the ablated variants. The proposed EfficientNet-B0 system reaches a top-1 accuracy of 73.1%, a top-5 accuracy of 92.8%, a macro precision of 74.5%, a macro recall of 73.2% and a macro F1-score of 73.2%, with a macro AUC of 0.995. It outperforms ResNet50 (70.4% top-1) and VGG16 (68.8%) on every accuracy metric in the table, although it has the fewest parameters of the three (5.6M against 25.9M and 15.4M). It is also by far the cheapest in arithmetic: EfficientNet-B0 needs 0.39 billion floating-point operations per image, compared with 4.1 billion for

TABLE III

RESULTS ON CUB-200-2011 (200 SPECIES). VAL. = TOP-1 ACCURACY ON THE VALIDATION SET USED FOR MODEL SELECTION; ALL OTHER COLUMNS ARE ON THE 5,794 TEST IMAGES. “FROZEN” IS THE TOP-1 ACCURACY AFTER STAGE 1 ONLY. P, R AND F1 ARE MACRO-AVERAGED. LATENCY IS MEASURED FOR ONE IMAGE AT A TIME ON A 2-CORE INTEL XEON AT 2.20GHZ CPU.

Model	Val. (%)	Frozen (%)	Top-1 (%)	Top-5 (%)	P (%)	R (%)	F1 (%)	AUC	Params	Size (MB)	Latency (ms)
VGG16 [2]	67.8	60.3	68.8	90.6	70.3	69.2	69.0	0.992	15.4M	59	609.4
ResNet50 [9]	72.8	62.3	70.4	91.4	71.5	70.7	70.6	0.993	25.9M	99	205.3
EfficientNet-B0 [10] (proposed)	75.2	69.6	73.1	92.8	74.5	73.2	73.2	0.995	5.6M	21	64.0
<i>Ablation: trained without data augmentation</i>											
VGG16	68.5	58.7	68.8	89.3	69.5	69.1	68.8	0.992	15.4M	59	609.4
EfficientNet-B0	75.2	73.2	74.8	93.0	75.3	75.0	74.8	0.994	5.6M	21	64.9

ResNet50 [10]. On the CPU it classifies an image in 64.0 ms, more than three times faster than ResNet50 (205.3 ms) and about nine times faster than VGG16 (609.4 ms). Even with its backbone frozen, EfficientNet-B0 (69.6%) is more accurate than the fine-tuned VGG16. The ranking on the test set agrees with the ranking on the validation set used for model selection.

C. Effect of Fine-Tuning and Data Augmentation

Fine-tuning. The “Frozen” column of Table III shows the accuracy reached with the backbone frozen. Fine-tuning the top of the backbone improves the top-1 accuracy of every model: by +3.5 percentage points for EfficientNet-B0, +8.1 for ResNet50 and +8.6 for VGG16. Adapting the high-level ImageNet features to birds is therefore essential in this fine-grained setting. The gain is largest for VGG16 and ResNet50, whose frozen features are weaker than those of EfficientNet-B0.

Data augmentation. Contrary to our expectation, the augmentation described in Section III-C did not improve accuracy. For VGG16, the model trained without augmentation reaches the same top-1 accuracy (68.8% against 68.8%), with a top-5 accuracy only 1.3 points lower. For EfficientNet-B0, the model trained without augmentation reaches 74.8% top-1 accuracy on the test set, 1.7 points more than the augmented model, although both reach the same validation accuracy (75.2%); the difference is already present after stage 1 (73.2% against 69.6%). We keep the augmented model as the proposed system because the design and the selection criterion were fixed before testing, and the validation set, with three images per species, cannot distinguish the two variants; we report the difference so that readers can weigh it. One possible explanation is that, with strong pretrained features and early stopping limiting training to a few epochs, the extra variability introduced by rotation and zoom slows the fitting of the head more than it improves generalisation. Augmentation designed for fine-grained recognition is left for future work.

D. Training Behaviour

Fig. 2 shows the accuracy and loss of the proposed model on the training and validation sets over both stages; the dashed line marks the start of fine-tuning. Training curves are computed on augmented images and with dropout active, while validation curves are not, so the two are not directly comparable in absolute terms. The widening gap between

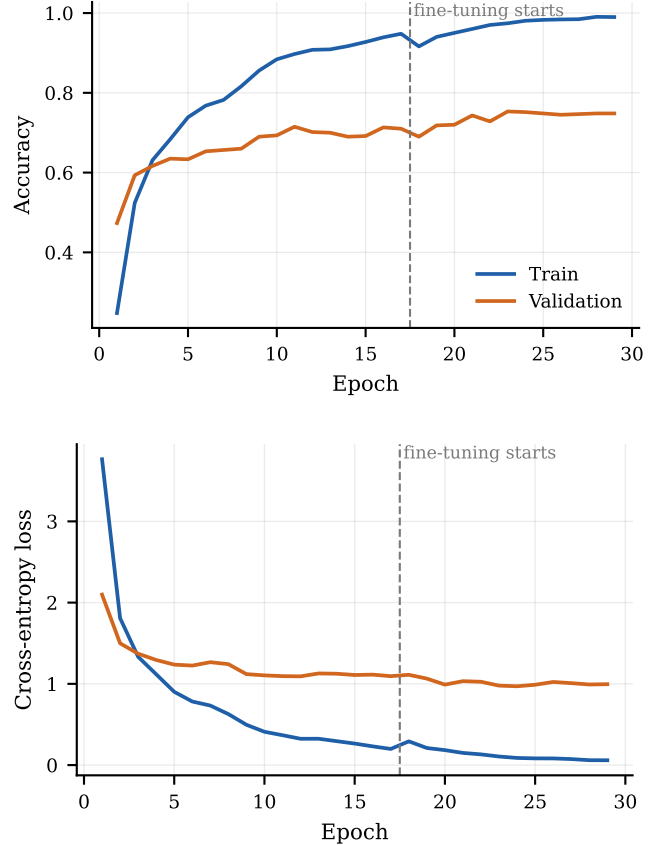


Fig. 2. Training and validation accuracy (top) and loss (bottom) of the proposed EfficientNet-B0 model. Epochs to the left of the dashed line belong to stage 1 (frozen backbone); those to the right belong to stage 2 (fine-tuning).

them, with training accuracy approaching 100%, shows that the model can memorise the roughly 27 training images per species; early stopping on the validation loss limits this overfitting.

E. ROC and Precision–Recall Analysis

Fig. 3 shows the micro-averaged ROC and precision–recall curves of the three fine-tuned backbones and of the frozen EfficientNet-B0, obtained by treating each of the $5,794 \times 200$ image–species pairs as a binary decision. The proposed model reaches a macro AUC of 0.995 and a mAP of 79.8%. The false

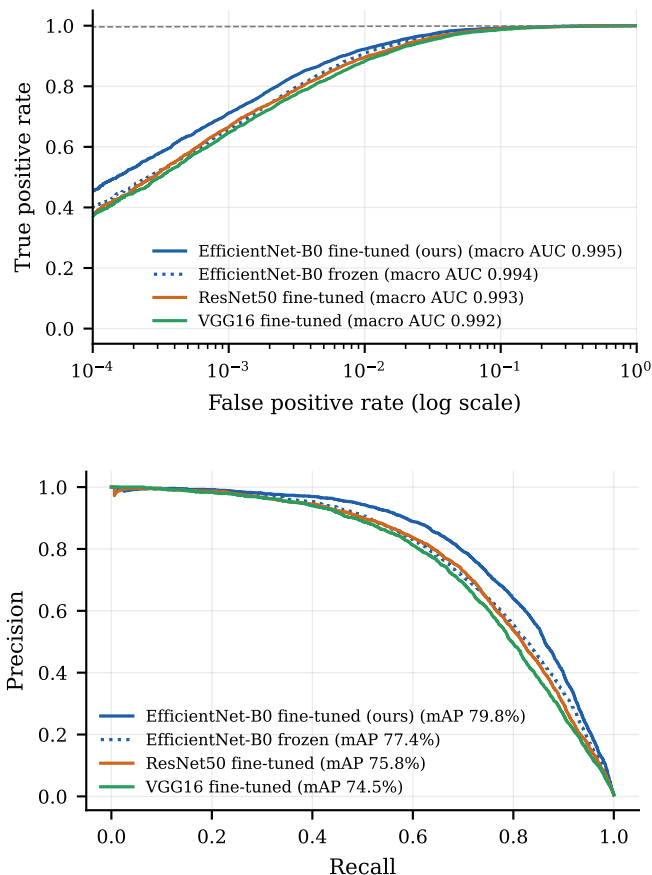


Fig. 3. Micro-averaged ROC curves (top, logarithmic false positive rate) and precision-recall curves (bottom) on the test set.

positive rate is plotted on a logarithmic scale because, with 200 classes, the operating points of interest lie at very low false positive rates.

F. Visual Explanations

Fig. 4 shows randomly chosen test images with their Grad-CAM heat maps for four correct and two incorrect predictions of the proposed model. In all six cases the highest activations lie on the bird, mostly on the head and upper body, rather than on the branches, water or vegetation in the background, which indicates that the model has learned features of the birds themselves. Both errors involve small, drably coloured songbirds with similar silhouettes: an Olive-sided Flycatcher predicted as a White-eyed Vireo and a Bank Swallow predicted as a Western Wood-Pewee. The first of these was predicted with a confidence of only 49%, which illustrates how the confidence score can flag uncertain identifications for review by a human expert.

G. Comparison with the Literature

To place these results in context, Table IV lists accuracies reported in the literature for methods designed specifically

TABLE IV
TOP-1 ACCURACY REPORTED IN THE LITERATURE ON CUB-200-2011
(NO BOUNDING BOXES AT TEST TIME)

Method	Backbone	Top-1 (%)
Part-based R-CNN [15]	AlexNet-based R-CNN	73.9
Bilinear CNN [16]	VGG-M + VGG16	84.1
RA-CNN [17]	VGG19	85.3
TransFG [18]	ViT-B/16	91.7
Proposed system	EfficientNet-B0	73.1

for fine-grained recognition. These methods use part localisation, attention or transformer backbones and, in several cases, higher input resolutions, so the comparison is indicative rather than controlled. They show the accuracy that specialised architectures can reach, whereas the proposed system uses a compact standard backbone at 224×224 pixels and a simple training procedure that is easy to reproduce and deploy.

H. Per-Species Analysis

Aggregate accuracy hides differences between species. Fig. 5 shows the distribution of per-species F1-scores of the proposed model. They range from 25.0% to 100.0% with a median of 76.0%, and 25 of the 200 species have an F1-score below 50%. Table V lists the five best- and worst-recognised species. The best-recognised species, such as the Parakeet Auklet, which is classified without a single error, and the White-breasted Kingfisher, have distinctive shapes or colour patterns. The worst-recognised species belong to groups that are notoriously difficult even for experienced birders: crows, terns, large gulls and the drab Western Wood-Pewee. Table VI confirms this: all six of the most frequent confusions are between closely related look-alike species, namely Common and Arctic Terns, Caspian and Elegant Terns (in both directions), Acadian and Least Flycatchers, and American Crow and Common Raven (in both directions). The model's errors are therefore concentrated where the visual differences are genuinely subtle, rather than spread at random.

Fig. 6 shows the full 200×200 confusion matrix, normalised by row. Because CUB-200-2011 numbers related species consecutively (for example, the gulls, terns, sparrows and warblers each form contiguous blocks), confusions between closely related, visually similar species appear as off-diagonal entries close to the diagonal, whereas confusions between unrelated species appear far from it.

I. Discussion and Limitations

The results show that, under an identical transfer-learning protocol, the choice of backbone matters: the compact EfficientNet-B0 is the most accurate, the smallest and the fastest of the three networks, which makes it well suited to a system intended for mobile or field use. Fine-tuning the top of the backbone helps every network, whereas our data augmentation did not. The top-5 accuracy, which is relevant when the system suggests a short list of candidate species to a human observer, is 92.8%, compared with 73.1% for top-1.

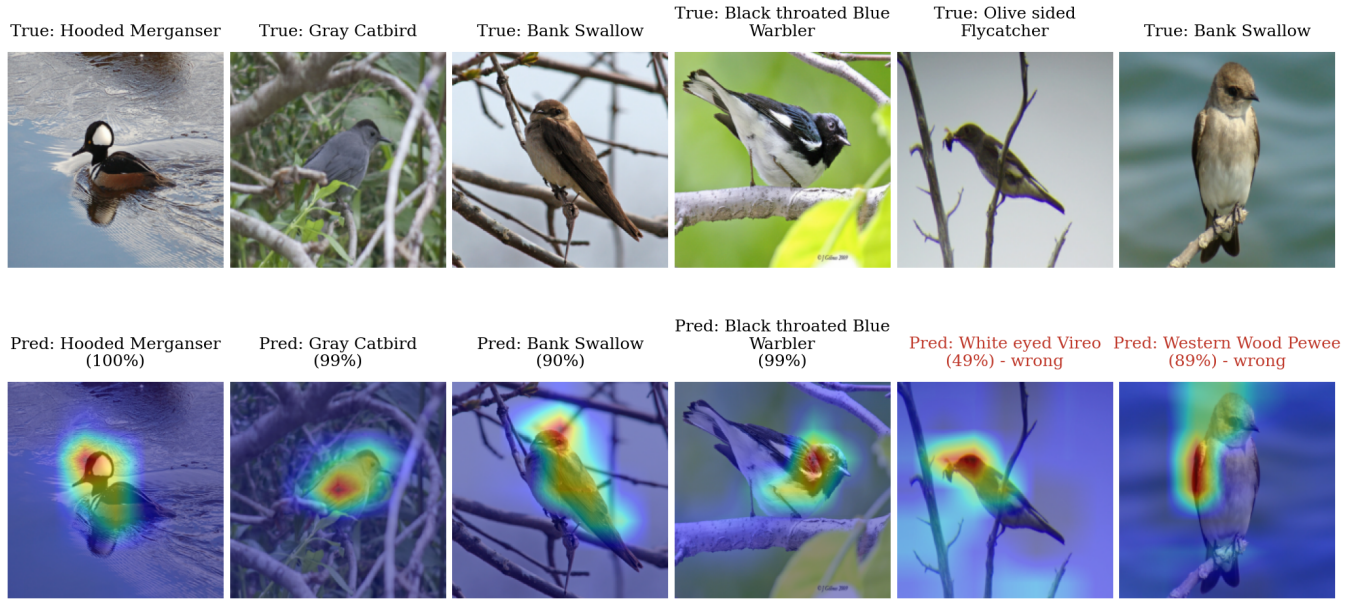


Fig. 4. Example outputs of the system. Top: test images with their true species. Bottom: predicted species, confidence and Grad-CAM heat map (red = regions that contributed most). The last two columns are misclassifications.

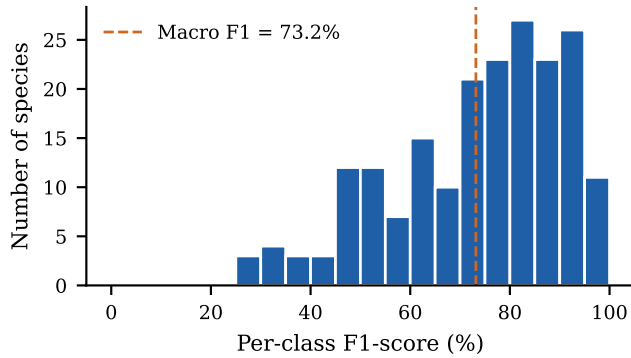


Fig. 5. Distribution of per-species F1-scores of the proposed model on the test set.

TABLE V
BEST- AND WORST-RECOGNISED SPECIES (PROPOSED MODEL, TEST SET;
 n = NUMBER OF TEST IMAGES)

Species	P (%)	R (%)	F1 (%)	n
<i>Five best-recognised species</i>				
Parakeet Auklet	100.0	100.0	100.0	23
Western Grebe	96.8	100.0	98.4	30
White-breasted Kingfisher	100.0	96.7	98.3	30
Green Jay	100.0	96.3	98.1	27
Heermann's Gull	93.8	100.0	96.8	30
<i>Five worst-recognised species</i>				
American Crow	23.5	26.7	25.0	30
Common Tern	31.8	23.3	26.9	30
Western Wood-Pewee	33.3	23.3	27.5	30
Herring Gull	50.0	23.3	31.8	30
Fish Crow	31.2	33.3	32.3	30

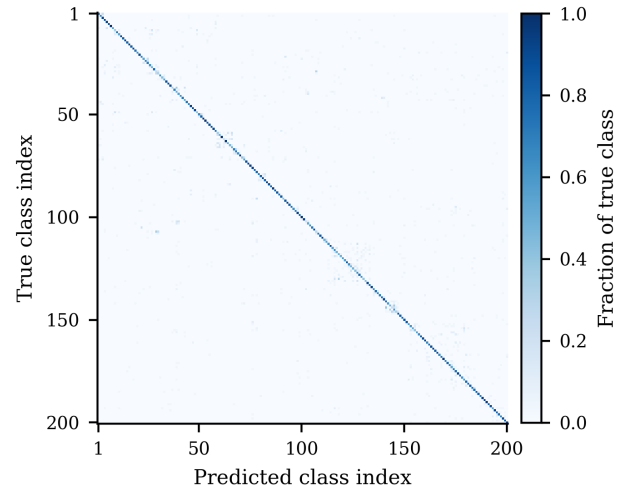


Fig. 6. Row-normalised confusion matrix of the proposed model over all 200 species (class indices follow the dataset's taxonomic ordering).

TABLE VI
MOST FREQUENTLY CONFUSED SPECIES PAIRS (PROPOSED MODEL, TEST SET)

True species	Predicted as	Count
Common Tern	Arctic Tern	12 of 30
Acadian Flycatcher	Least Flycatcher	11 of 29
Caspian Tern	Elegant Tern	11 of 30
Common Raven	American Crow	10 of 30
Elegant Tern	Caspian Tern	10 of 30
American Crow	Common Raven	9 of 30

Several limitations remain. First, CUB-200-2011 contains only North American species photographed mostly in good conditions; performance on camera-trap or smartphone images, and on regional species such as those of India, remains to be verified. Second, the dataset provides about 30 training images per species, and the training curves show clear overfitting despite augmentation, dropout and early stopping. Third, the validation set holds only three images per species, which is too few to separate closely matched variants such as the models with and without augmentation. Fourth, specialised fine-grained architectures still achieve higher accuracy than our system (Table IV), at the cost of greater complexity. Finally, the latencies in Table III were measured on a 2-core Intel Xeon at 2.20GHz CPU; measurements on phones would be needed to confirm the deployment advantage of EfficientNet-B0.

V. CONCLUSION AND FUTURE WORK

We presented a CNN-based bird species identification system that combines an ImageNet-pretrained EfficientNet-B0 backbone with a regularised classification head and two-stage transfer learning. Evaluated on the official test split of CUB-200-2011 across all 200 species, it achieves a top-1 accuracy of 73.1%, a top-5 accuracy of 92.8% and a macro F1-score of 73.2%, outperforming VGG16 and ResNet50 trained under the same protocol while using the fewest parameters and running fastest. Fine-tuning improved every backbone, while the data augmentation used did not improve accuracy. The per-species analysis shows that the remaining errors are concentrated among closely related look-alike species, and Grad-CAM confirms that the predictions are based on the birds rather than the background.

In future work we plan to (i) add attention or part-localisation mechanisms to close the gap with specialised fine-grained methods, (ii) evaluate the system on larger and more diverse datasets such as NABirds and iNaturalist and on regional species, and (iii) combine image-based and acoustic identification.

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